

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

WIRELESS SENSOR NETWORK

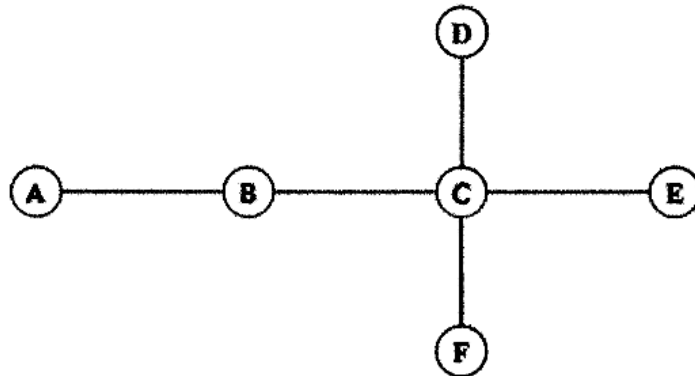
Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as mentioned.
 - (2) Draw neat and clean diagram, wherever necessary
 - (3) Use of Non – programmable calculator is permitted.
 - (4) Assume suitable data wherever necessary.
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1. (a) Explain the principle of piezoelectric sensor to measure movement.
How can a magnetic sensor be employed to measure the movement of vehicles ? 5(CO5)
 - (b) Compare the difference between event-based and thread-based operating systems.
Discuss some of the advantages and disadvantages of the two approaches in the context of wireless sensor networks. 5(CO1)
 2. (a) Discuss sensor node components and demonstrate architecture of Hightrob sensor node with a neat sketch. 10(CO1)
 3. (a) Evaluate the trade-offs between power consumption, performance and design complexity in ultra-low power sensor node design. 5(CO5)
 - (b) What is the "early sleeping problem" and how does T-MAC address this problem ? 5(CO3)
 4. (a) Assume that the RTS and CTS frames were as long as the DATA and ACK frames. Would there be any advantage to using the RTS/CTS approach ? Explain why or why not. 5(CO3)
 - (b) What is the main purpose of the MAC layer and why is this challenging in networks with shared media ? 5(CO2)

5. (a) In a synchronized network with unknown synchronization error, an anchor node periodically broadcasts an acoustic signal to sensor nodes in its range. At time 1000 ms on the anchor node's clock, the anchor node issues a beacon, which is received by node A at time 2000 ms (on node A's clock). What is the distance that A can now compute ? 5(CO4)
- (b) Instead of computing the distance itself, node A also responds with an acoustic signal issued at time 2500 ms, which is received by the anchor node at time 3300 ms. What is the distance computed by the anchor node ? What can you say about the synchronization of anchor node and node A ? 5(CO4)
6. (a) Consider the network topology in Figure, where the lines indicate which nodes can communicate and interfere with each other. Assume a TDMA protocol with a frame size of 5 slots and that each node can only be sender or receiver during any time slot.



- (a) Generate a schedule such that every node has an opportunity to communicate to all of its neighbors.
- (b) For your schedule, how many slots in a frame could each node sleep to preserve energy ? What is your insight with respect to node density and energy preservation ?

Assume that node A sends a message to node E ; how long (in number of time slots) does it take for E to receive the message using your schedule ? (Explain your answer.) 5(CO3)

- (b) What is the difference between a proactive routing protocol and a reactive routing protocol ? Explain at least two examples for each category. 5(CO2)

